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VANGUARD

A/B TEST ANALYSIS

2017

Did the new UI improve client completion rates?

A data-driven analysis of Vanguard's digital UX experiment

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Introduction

About Vanguard

Vanguard is one of the world's largest investment management companies, serving over 50 million investors globally.

As part of a digital transformation initiative, the UX team designed a new online process with a refreshed interface and in-context prompts to guide client's step by step.

The goal: reduce friction and increase the number of clients completing the process.

The Central Question

Did the new UI lead to a statistically significant improvement in client completion rates?

Test period: March 15 – June 20, 2017

Groups: Control (old UI) vs Test (new UI)

Success threshold: $\geq 5\%$ improvement

Data Overview & Preparation

01 Client Profiles

df_final_demo

Demographics: age, gender, tenure, balance

02 Digital Footprints

df_final_web_data (pt_1 + pt_2)

Step-by-step interactions with timestamps

03 Experiment Roster

df_final_experiment_clients

Client assignment: Control or Test group

Data Cleaning

1 Merge

Combined pt_1 + pt_2, joined with demo and experiment tables

2 Nulls

126,662 rows without group assignment identified (demo-only)

3 Types

date_time → datetime64; process steps mapped to numeric order

4 Outliers

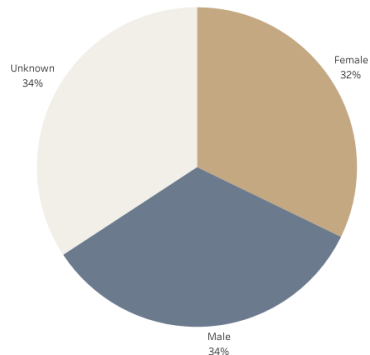
Sessions > 30 min filtered out as inactive

5 Features

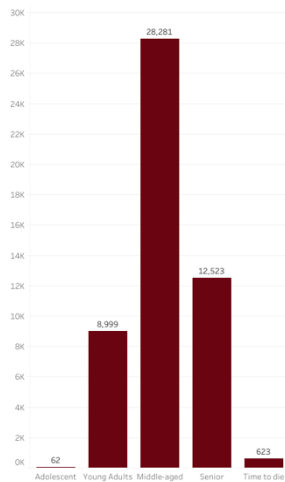
Added step_num, time_spent, prev_step_num, is_error

Who Are Vanguard's Clients?

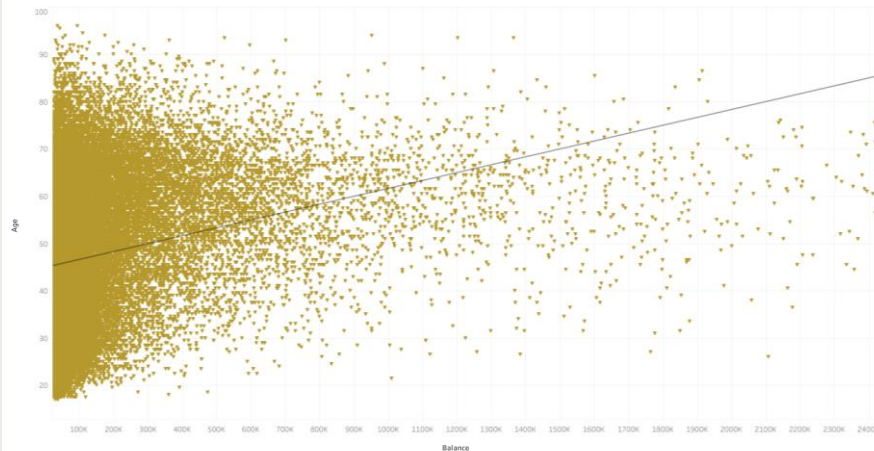
Gender Distribution



Age Groups



Age vs Balance



Key insight: most clients are middle-aged (31–59), with balance increasing with age. Gender is evenly split across Male, Female and Unknown, which suggests incomplete demographic data.

KPI 1 — Completion Rate

Definition: % of unique visits that reach 'confirm' from 'start' · Success: Test > Control by ≥5%

Control

51.91%

completion rate

Test

65.54%

completion rate

Difference

+13.63%

 beats 5% threshold

KPI 2 — Average Time Spent per Step

Definition: Mean duration (seconds) per step per visit · Success: Test < Control (faster = better UX)

Step	Control (sec)	Test (sec)	Faster?
Start	60.91	56.24	✓ Test
Step 1	47.46	58.48	✗ Control
Step 2	90.21	88.13	✓ Test
Step 3	133.30	124.83	✓ Test
Confirm	156.16	221.59	✗ Control

Key insight: test group was significantly faster at Start, Step 2 and Step 3. No significant difference at Step 1. Test users spent considerably more time at Confirm (a negative finding)

KPI 3 — Error Rate

Definition: % of visits where users moved backward from a step · Success: Test < Control (less friction)

Step	Control error %	Test error %	Diff	Result
Start	0.00%	0.00%	0.00%	—
Step 1	9.57%	18.04%	+8.47%	✗ Test worse
Step 2	8.91%	16.17%	+7.26%	✗ Test worse
Step 3	20.94%	19.78%	-1.16%	✓ Test better
Confirm	39.03%	9.61%	-29.42%	✓ Test better

Interesting finding: test group had higher error rates at Steps 1–2, but a dramatically lower rate at Confirm (9.61% vs 39.03%). The new UI did not reduce overall friction

Hypothesis Testing

01 Completion Rate

H₀ Completion rate (Test) $\leq$ Completion rate (Control)

H₁ Completion rate (Test) $>$ Completion rate (Control)

 *Reject H₀*
 $Z = 35.04 \cdot p < 0.000001$

Test: 65.54% vs Control: 51.91%
The new design significantly improved process completion

02 Time Spent per Step

H₀ Test group time $\geq$ Control group time

H₁ Test group time $<$ Control group time

 *Reject H₀ at Start, Step 2, Step 3*
 *Fail to reject H₀ at Step 1 and Confirm*

Start ($p=0.0011$) · Step 2 ($p=0.0101$) · Step 3 ($p=0.0001$): test group significantly faster

Step 1 ($p=1.0$): no significant difference

Confirm ($p=0.9999$): test group is actually slower (negative finding)

03 Error Rate

H₀ Error rate (Test) $\geq$ Error rate (Control)

H₁ Error rate (Test) $<$ Error rate (Control)

 *Fail to Reject H₀*
 $Z = 7.9996 \cdot p = 1.0000$

The new UI did not reduce errors: test group showed higher error rate than control

Experiment Evaluation

Design Effectiveness

Well-structured control/test split; unassigned client IDs were dropped to ensure clean group comparison.

Randomisation & Balance

Groups unequal (~3k more in test), assignment method undocumented; bias expected but not seen in visualisations.

Demographic Balance

No meaningful bias detected across age, gender, tenure, balance or account count.

Duration

3-month window (Mar–Jun 2017) was adequate, though seasonal effects cannot be fully ruled out.

Additional Data Needs

Clearer step-flow mapping, app context, and client goals would enable more targeted recommendations.

Tableau Dashboard

Live Dashboard

[\[Link \]](#)

Teamwork & Learnings

O1 Team Management

Week 1: KPIs and Hypothesis Testing divided individually (one task per person).

Week 2: One big task per team member, with shared ownership of deliverables.

Leadership shifted from Week 1 (assigned lead) to Week 2 (shared knowledge and peer training).

*Tools: Trello for task tracking · GitHub for version control
· at least 1 commit per project day.*

O2 Challenges & Learnings

Challenges: Communication: coordinating decisions, naming conventions, code structure and which dataset to use at each stage of the analysis.

Building one beautiful, consistent, streamlined dashboard that works 100%.

Learnings: Regular review is essential to ensure consistency in grouping methodology. Being precise and strict with the questions we ask, and our answers, is key to avoiding misaligned results.

Different datasets do not allow filtering of visualizations built with them.

Strengths, Limitations & Next Steps

01 Strengths

- Large sample size: 317,235 rows, 30k+ unique visits per group
- Real user data from a genuine A/B experiment with proper control/test split
- Multiple KPIs analysed across completion, time, errors and funnel drop-off (drop-off was measured but ultimately excluded from the formal analysis)
- No meaningful demographic bias confirmed through Tableau visualizations

02 Limitations

- 126,662 rows had no group assignment and were excluded
- Possible incomplete demographics data
- Test period only ~3 months (seasonal effects cannot be ruled out)
- Assignment method undocumented (minor randomisation risk remains)
- Step-flow logic unclear: uncertain which steps can lead backwards to which

03 Next Steps

- Investigate why users go backwards at the start of the process
- Re-run analysis with demographic filters to check if specific segments benefit more
- Extend or repeat the test across different seasons to confirm results hold

Conclusion

Final Verdict

The new UI significantly boosted completion rates: up 13.63%, well above the 5% target. Users in the test group dropped off less at the beginning and end of the funnel and progressed further through the process.

- Completion rate jumped from 51.91% to 65.54%, meaning the new design works.
- Time improvements were significant at Start, Step 2 and Step 3; but test users were actually slower at Confirm.
- Error rates were higher in the test group; users navigated backwards more, especially early on.
- Recommendation: roll out the new UI, but investigate why users are confused at the start of the process.



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Thank you

Questions?

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